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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Chanel
Selected Domain	Domain 1
Selected SDG Track (1–17)	4, 5, 8, 9, 10
Project / Solution Name	KompasAI
Name (Leader)	Brayden Chong Jie Rui
Email & Phone (Leader)	braydencjr05@gmail.com
University Affiliation(s)	Universiti Malaya
Number of Team Members	2

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is reshaping work faster than any person can track. New models, tools and skills appear every week across research papers, open-source projects and job postings, and no student or working adult can monitor that firehose around the clock while still finding time to grow. Picture an accounts executive who keeps hearing "AI will take your job" but has no idea which daily tasks are actually at risk, or a final-year student scrolling through endless "must-learn" lists without knowing which skills local employers will really hire for. Most of what they see is hype, and they cannot tell a passing trend from real Malaysian job demand. Career advice offers little help: it is a one-off test or a static list that is outdated by the time it is published. So people guess. They spend months and savings on courses that lead nowhere, commit to paths they later discover they don't enjoy, or freeze and do nothing until change catches them unprepared. The burden falls hardest on those with the least spare time and guidance, such as mid-career workers in routine roles and graduates or career switchers without mentors, widening inequality. Meanwhile, emerging roles in areas like ESG reporting and AI governance stay invisible and understaffed, slowing progress on decent work and climate goals. People don't need more information; they need the market watched for them automatically, hype filtered from real demand, and every shift turned into their next step. AI watches the market. You grow.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Kompas is an always-on career signal engine: AI watches the market so users can focus on growing. At its core is an automatic ingestion pipeline that continuously collects signals from AI research (arXiv), open-source activity (GitHub and Hacker News) and Malaysian job demand (Department of Statistics Malaysia Job Market Insights and TalentCorp's Critical

Occupations List). Each signal is tagged to skills from the ESCO taxonomy, and every skill climbs a Signal Ladder from research buzz to hiring abroad to real Malaysian job demand, earning a status such as Watch, Emerging, Act now or Declining. Fixed rules over this data set every score, which separates hype from genuine opportunity. An AI Live Trend page adds Malaysian news, Reddit and trending GitHub projects, and tells users what moved since the last snapshot. For the user, the experience is simple. They upload a resume, and Google's Gemini reads it into a skills profile, quoting the resume word for word so nothing is invented. Using the International Labour Organization's generative-AI exposure index, Kompas shows which of their tasks AI is reaching and which still need them, and how their skills hold up by 2028. Career Impact then offers three kinds of option: Stay and strengthen, Fastest-growing and Safest switch, each with a match score, skills gap, SDG link, learning route and live job openings. Gemini writes the reasons from cited evidence and marks any estimate. Before committing, users can try a role in a work simulation generated by Anthropic's Claude, so they find out whether they enjoy the work. We apply skills from the SDC Phase 1 workshops: GenAI APIs (Gemini and Claude with structured outputs), database architecture (Supabase Postgres storing raw records, signals and dated snapshots) and cloud hosting (the Next.js frontend and FastAPI backend deployed on Replit).

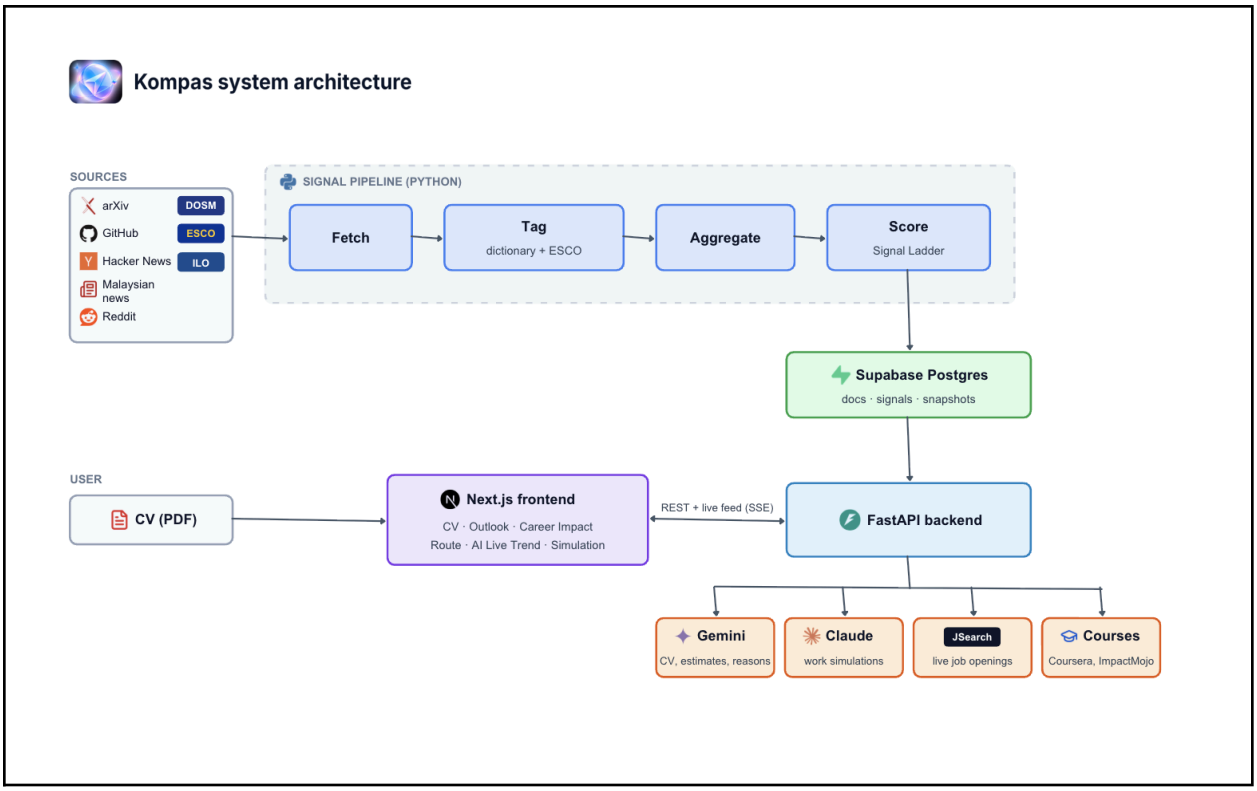
3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- **Frontend:** Next.js 15, React 19, TypeScript and Tailwind CSS 4, with a live feed over server-sent events and Sign in with Google
- **Backend**: Python with FastAPI (uvicorn), httpx, pypdf for PDF resumes, and openpyxl for Department of Statistics Malaysia spreadsheets
- **GenAI APIs:** Google Gemini 2.5 Flash (google-genai) turns resumes into structured skill profiles, with verbatim quotes checked in code. It also writes the reasons behind each option from cited evidence, estimates values where data is missing (marked as estimates), tags SDGs and summarises AI Live Trend headlines. Gemini Embedding matches skills by meaning. Anthropic Claude (anthropic SDK) generates the work simulations and grades the "direct the AI" task.
- **Data sources and APIs:** arXiv API, GitHub Search API and GitHub Trending, Hacker News (Algolia) API, Department of Statistics Malaysia Job Market Insights, TalentCorp's Critical Occupations List (MyCOL), the International Labour Organization's generative-AI exposure index, the ESCO API for the skills taxonomy, Malaysian news RSS feeds and Reddit for AI Live Trend, the JSearch API for live Malaysian job openings, and Tavily, Coursera and ImpactMojo for course suggestions
- **Database:** Supabase Postgres (psycopg with a connection pool), with SQLite for offline or local use. It stores raw records, tagged evidence, Signal Ladder scores and dated snapshots.
- **Hosting:** Replit Deployments runs the frontend and backend together at https://kompas-ai.replit.app, with API keys in Replit Secrets.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We followed the five design-thinking stages.

Empathise: We built personas around real moments of need. We often hear people around us , or online , especially fresh graduates facing difficulties when they are seeking for jobs , as they are unsure about trend , and often being "brain-washed" with posting of "AI is replacing human job". However, we believe in AI is here to help and make things easier, not replacing human and humanity.

Define: The problem is not too little information but too much, so the core need became a system that watches signals for users and filters hype from real demand.

Ideate: We mapped one journey (your work today, your skills and AI outlook, career impact caused by AI, your route, AI live trends) so each step answers one question.

Prototype: we built a working Next.js and FastAPI app with a live signal pipeline, Gemini CV reading and Claude work simulations.

Test: Automated tests cover connectors, scoring and CV reading, while persona walkthroughs check whether recommendations are clear. CV skills are verified against direct quotes, and simulations are checked against reference solutions. Stored data provides a fallback if a source fails

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	● **4.4:** "Substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Kompas gives each user a personalised learning route toward skills with proven Malaysian job demand. ● **5.b:** "Enhance the use of enabling technology, in particular information and communications technology, to promote the empowerment of women." Kompas is a free AI career guide that is especially useful to women returning to work or switching careers. ● **8.5:** "By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value." Kompas matches users to roles that fit their skills, and its work simulation lets them try a job before committing to it. ● **9.5:** "Enhance scientific research, upgrade the technological capabilities of industrial sectors in all countries, in particular developing countries…" Kompas follows AI research and open-source activity and turns it into the skills Malaysian industry is starting to hire for. ● **10.2:** "By 2030, empower and promote the social, economic and political inclusion of all, irrespective of age, sex, disability, race, ethnicity, origin,

Metric Category	Your Response
	religion or economic or other status." The CV reader never extracts age, gender, ethnicity, nationality or religion, so recommendations can't depend on them.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	● **AI skills (4.4, 9.5):** 56% of Malaysian professionals use AI tools at work, but only 26% have received formal training [2]. ● **Skills mismatch (8.5):** 35.6% of employed workers with tertiary education were in skill-related underemployment in Q2 2025 [3]. ● **Gender gap (5.b):** in Q4 2025, women's labour force participation was 56.6% and men's was 83.3%, a gap of 26.7 percentage points [4]. ● **Inclusion (10.2):** existing career tools give no transparent proof that their recommendations ignore personal characteristics. Kompas will set its own baseline using a counterfactual test.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. The share of pilot users who start the first step of their Kompas route within 7 days of their first session (SDGs 4.4 and 8.5). 2. The share of recommended skills backed by Malaysian job-demand evidence, meaning the skill reached the top rung of the Signal Ladder from DOSM or MyCOL data (SDG 9.5). 3. Track women's participation in the pilot and evaluate whether career roadmap start rates differ by gender, using voluntary self-reported data where available. 4. Recommendation changes when only the name, gender or age in a CV is changed (SDG 10.2).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation*	● At least 50% of 30 or more pilot users start their first route step within 7 days, roughly double the 26% formal-training baseline.

Metric Category	**Your Response**
e.g., "20% improvement in computational resource use."	• At least 80% of recommended skills are backed by Malaysian demand evidence. • Ensure equal access to career guidance and learning opportunities for all users, regardless of gender. • Changing only a name, gender or age in the counterfactual test changes 0% of recommendations.
Sources	1. United Nations, SDG indicators metadata (Targets 4.4, 5.b, 8.5, 9.5 and 10.2): https://metadata.un.org/sdg/ 2. Hays Asia Salary Guide 2025, as reported by People Matters: https://sea.peoplemattersglobal.com/article/strategic-hr/ai-in-malaysian-workplaces-a-strategic-outlook-46436 3. Department of Statistics Malaysia, Labour Market Review Q2 2025: https://www.dosm.gov.my/portal-main/release-content/labour-market-review-lmr-second-quarter-2025 4. Department of Statistics Malaysia, Labour Force Statistics, December and Q4 2025 (released 11 February 2026), as reported by RinggitPlus: https://ringgitplus.com/en/blog/personal-finance-news/dosm-female-labour-force-participation-hits-record-56-6-in-q4-2025.html

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✅ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✅ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type	License
Next.js 15	Code (framework)	MIT
React 19 / React DOM	Code (library)	MIT
TypeScript	Code (language/compiler)	Apache-2.0
Geist and Geist Mono fonts (via next/font)	Code (font)	SIL Open Font License 1.1

FastAPI	Code (framework)	MIT
Uvicorn	Code (server)	BSD-3-Clause
HTTPX	Code (library)	BSD-3-Clause
pypdf	Code (library)	BSD-3-Clause
openpyxl	Code (library)	MIT
psycopg 3 and psycopg-pool	Code (library)	LGPL-3.0
python-multipart	Code (library)	Apache-2.0
python-dotenv	Code (library)	BSD-3-Clause
Anthropic Python SDK	Code (library)	Apache-2.0
Google Gen AI SDK (google-genai)	Code (library)	Apache-2.0
pytest	Code (testing)	MIT
Claude (Anthropic API)	Model (pre-trained LLM)	Proprietary; Anthropic Commercial Terms of Service and Usage Policy

Gemini 2.5 Flash (Google Gemini API)	Model (pre-trained LLM)	Proprietary; Gemini API Terms of Service
ESCO skills and occupations (European Commission)	Dataset/API	CC BY 4.0
ISCO-08 occupation classification (ILO)	Dataset (classification)	ILO terms, free reuse with attribution
arXiv API metadata	Dataset/API	CC0 1.0 (metadata)
GitHub REST API (repository search)	Dataset/API	GitHub Terms of Service
Hacker News API (via Algolia)	Dataset/API	Public API; Y Combinator / Hacker News terms
DOSM Job Market Insights	Dataset	DOSM open data terms, with attribution
MyCOL Critical Occupations List (TalentCorp)	Dataset	Public government publication, with attribution
JSearch API (OpenWeb Ninja / RapidAPI)	Dataset/API	Commercial API Terms of Service (free tier)

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✅ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Date: 24/10/2026